

FREEFINE

Reproduction, metric-driven diagnosis, and a systematic architectural study on GeoBench-2D

Faithfully reproducing a training-free geometric image editor, locating its weakness objectively, and testing five distinct experiments against it.

What I will walk through

I

The Baseline

Task, decoupled pipeline, the TCA refinement engine, benchmark, metrics & compute.

II

Reproduction & Diagnosis

Bit-faithful Phase 0 reproduction, then a metric-driven diagnosis that localizes the weakness.

III

Five Experiments

Tuning, input quality, identity conditioning, geometry-aware TCA, and refinement preservation — full data.

IV

Synthesis & Direction

What the five experiments establish together, the rigor behind it, and the next directions.

PART I

The Baseline

What 2D geometric image editing is, the decoupled FreeFine pipeline, the training-free refinement engine, and the benchmark and compute it runs on.

Move, rotate, resize — within the image plane

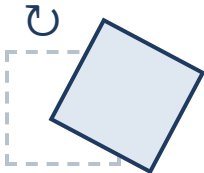
Every GeoBench-2D edit applies a known 2D affine transform to one object. The object is resampled to its new pose, the vacated region is inpainted, and the result must stay coherent.

MOVE



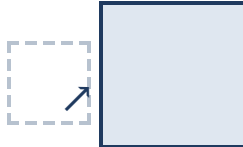
Pure translation. Content is preserved; only its position changes.

ROTATE



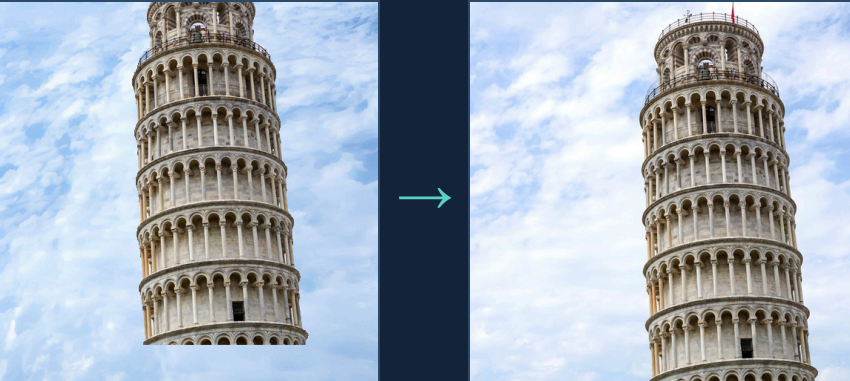
In-plane rotation. Object pixels are resampled; gaps appear at the corners.

RESIZE



Scaling. Object is interpolated up/down; surrounding area must adapt.

A REAL 2D EDIT

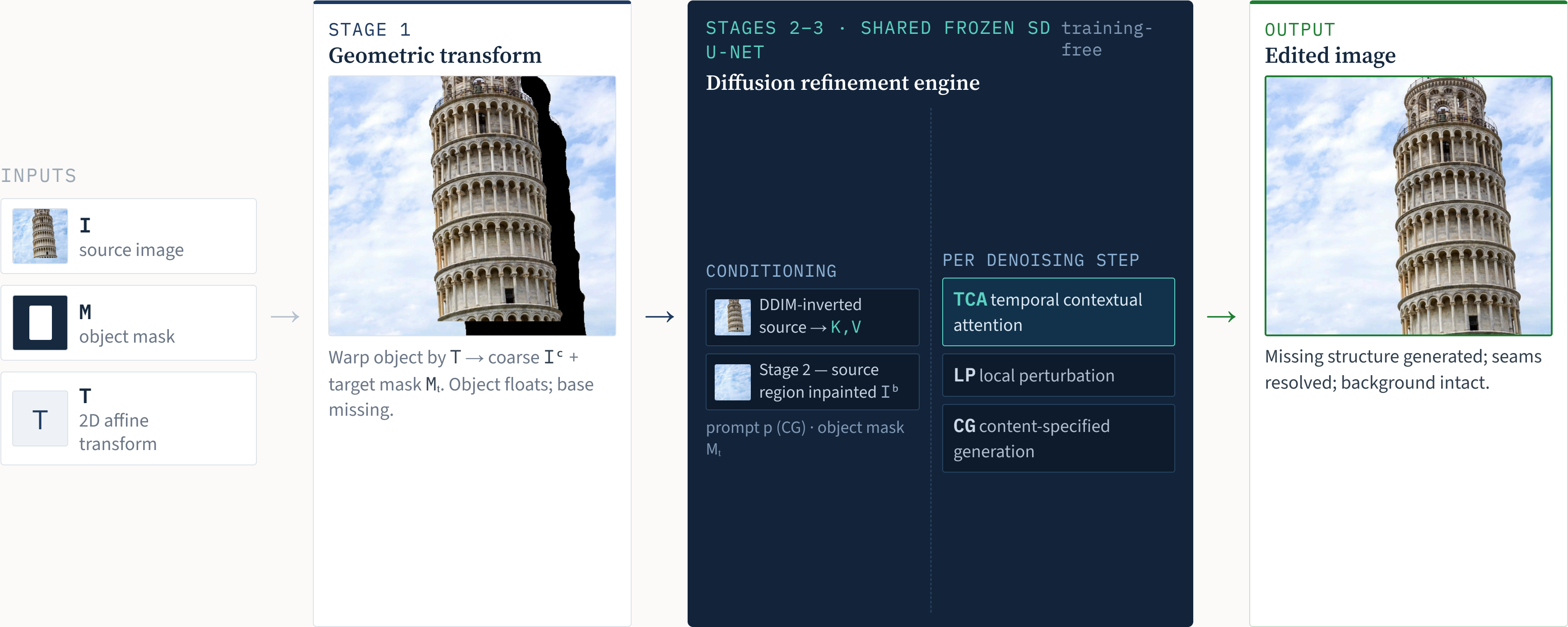


The tower is translated — a genuine in-plane edit.

Scope: out-of-plane / 3D edits that reveal genuinely hidden surfaces (GeoBench-3D) are **not** studied here — they are future **Direction A**.

A decoupled, training-free editing pipeline

Inputs are transformed by the known 2D affine, then a single frozen Stable-Diffusion U-Net — shared by inpainting and refinement — completes the edit. No step is trained.



Temporal Contextual Attention

The structural backbone of refinement: it carries the source object's geometry into the edited image, denoising step by step. This is the attention Experiment 4 targets.

TEMPORAL CONTEXTUAL ATTENTION (TCA)

In the denoising loop, the edit-image query attends to keys/values from the DDIM-inverted source. A weight α_τ falls 1→0 across steps: start on mask-guided mutual attention (global structure), end on plain self-attention (local detail).

$$\mathbf{f}_g^\tau = (1-\alpha_\tau) \cdot S_t + \alpha_\tau \cdot [S_o \cdot M_t + S_b \cdot (1-M_t)]$$
$$\alpha_\tau = (\tau_1 - \tau) / (\tau_1 - \tau_0) \quad \cdot \quad S = \text{softmax attention}, M_t = \text{object mask}$$



WHAT IT DOES

Reuses the source's own attention features so the edited object keeps the original's shape, texture and identity — no retraining, no extra network.

THE α SCHEDULE

Front-loads structure (high α early), then releases to detail ($\alpha \rightarrow 0$ late) — a controlled hand-off from *copying the source* to *refining the edit*.

The one gap: TCA matches source↔edit tokens by *feature similarity*, never by the *known* affine geometry. That blind spot is exactly what Experiment 4 targets.

Local Perturbation & Content-specified Generation

LOCAL PERTURBATION (LP)

At the start of refinement, re-noise the latent **only inside the mask M**; outside keeps its DDIM-inverted value. The injected noise is the model's freedom to synthesize — confined to the edit region.

$$z_t = M \odot (\sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{(1-\bar{\alpha}_t)} \cdot \varepsilon)$$
$$+ (1-M) \odot z_t^{inv}$$

$\varepsilon \sim \mathcal{N}(0, I)$ · M = object mask · z^{inv} = inverted latent

→ substantial change inside M, the rest pinned to the original.

CONTENT-SPECIFIED GENERATION (CG)

Classifier-free guidance on a prompt **p** that names the object. The prompt acts through the U-Net's **cross-attention**; the scale s sets how strongly it steers newly generated content.

$$\varepsilon_{\hat{\theta}}(z_t, p) = \varepsilon_{\theta}(z_t, \emptyset)$$
$$+ s \cdot (\varepsilon_{\theta}(z_t, p) - \varepsilon_{\theta}(z_t, \emptyset))$$

s = guidance scale · $\emptyset$ = null prompt · p via cross-attention

→ semantically correct completion of revealed / ambiguous areas.

STAGE 1 • TRANSFORM I^c , M_t = AffineWarp($I, M; \theta$) — warp the object by edit params θ .

STAGE 2 • INPAINT vacated source region filled by the **same frozen U-Net** (no extra model).

THE BENCHMARK

GeoBench-2D

The evaluation set: single-object geometric edits with known masks and exact transform parameters, stratified by operation and difficulty.

609

source images

5,677

single-type edits

3

operation families

EDITS BY OPERATION

1,439

move

1,603

rotate

2,635

resize

balanced-200

Reproducible stratified **seed-42** dev subset for cheap parameter sweeps — preserves move/rotate/resize proportions.

hard groups

The large-transform **rotate_hard (n=22)** and **resize_hard (n=23)** slices isolate where the pipeline is actually stressed.

Each case ships with **source image + object mask + exact edit parameters** — so the target geometry is known per case, and edits are stratified by transform magnitude (small vs large).

What each number actually measures

REALISM ↓ distribution distance	CONSISTENCY ↑ feature cosine	PRECISION ↓ geometric error
FID ↓ $\ \mu_r - \mu_g\ ^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{\frac{1}{2}})$ Fréchet distance between Inception feature Gaussians of real vs generated.	SUBC ↑ · headline $\text{mean } \cos(\phi(\text{obj}_{\text{edit}}), \phi(\text{obj}_{\text{src}}))$ Subject structural consistency — did the object keep its structure?	WE / WRAP_E ↓ $\sum I^c \cdot m - I_{\text{gen}} \cdot m / \sum m$ Warping error — L1 of expected-warp vs output over the object mask.
DINOv2 / FDD ↓ $\text{Fréchet}(\phi_{\text{DINOv2}}(X_r), \phi_{\text{DINOv2}}(X_g))$ Same form on DINOv2 features — more semantic than Inception.	BGC ↑ $\text{mean } \cos(\phi(\text{bg}_{\text{out}}), \phi(\text{bg}_{\text{in}}))$ Background consistency — scene outside the edit left untouched.	MD ↓ · headline $\text{mean } \parallel \text{kp}_{\text{src}} - \text{kp}_{\text{gen}} \parallel \quad (\text{DIFT})$ Mean keypoint distance via DIFT (SD-2.1) correspondences — how far keypoints drifted. The one stochastic metric.
KD ↓ $\text{MMD}^2(X_r, X_g) \cdot \text{poly-kernel}$ Kernel distance (KID) — unbiased, robust on small samples.		

PART II

Reproduction & Diagnosis

First confirm I match the paper bit-for-bit. Then locate the weakness objectively, by difficulty and edit type — before touching the method.

All seven metrics reproduce within ~2%

Metric	Paper	Reproduced	Diff
FID ↓	34.72	35.04	+0.9%
DINOv2 / FDD ↓	478.18	487.82	+2.0%
KD ↓	0.144	0.142	−1.2%
SUBC ↑	0.907	0.911	+0.4%
BGC ↑	0.971	0.967	−0.4%
WE ↓	0.055	0.047	−14.5%
MD ↓	9.25	8.50	−8.1%

5,677 edits generated & scored, fp32 / seed 42 · bit-identical generation confirmed.

WHY WE & MD EDGE THE PAPER

Not metric handling — Phase 0 uses the **official, repo-default** metrics. The small gains (WE 0.047 vs 0.055; MD 8.50 vs 9.25) are attributable to **full-precision fp32 inference on TF32-free hardware (T4)** and newer CUDA kernels, seed & model held at repository values.

PROTOCOL CAVEAT

This comparison is valid. The **ablations (Exp 1–5)** use seeded-MD + a direct WE re-implementation — internally consistent vs our *own* baseline, but not directly comparable to the paper's WE/MD.

Five changes to run it — none touches the algorithm

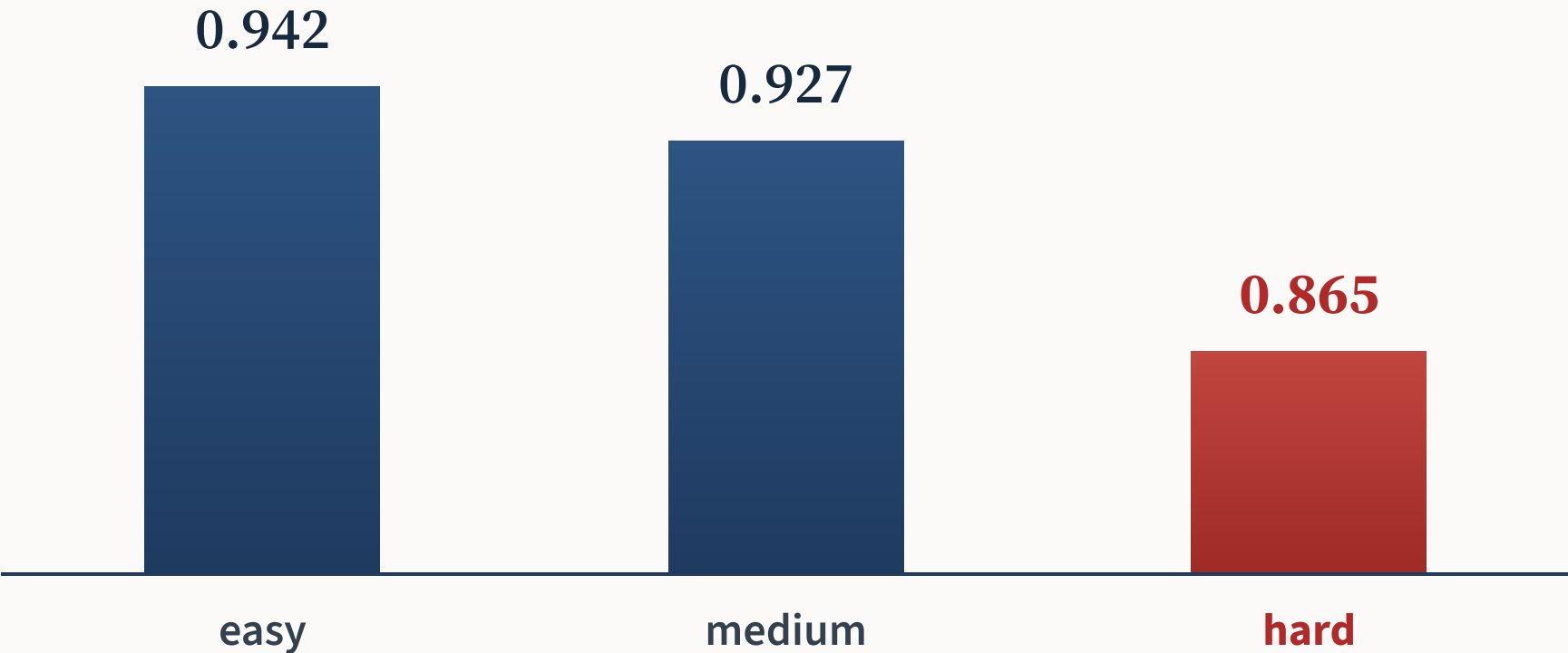
Two just re-point at identical weight mirrors, two fix real crash bugs in the released code, one pins library versions. Each was checked to leave results unchanged.

WEIGHT MIRROR neutral	The model weights moved. The original SD-1.5 & SD-2.1 download pages were removed, so I point at community mirrors. They are bit-for-bit identical — generation and the MD metric are unaffected.
CRASH BUG code fix	A mask was used before it was loaded. <code>ori_mask</code> was read by the warp before being assigned. I reordered the two lines so the intended mask loads first — same mask, no crash.
CRASH BUG code fix	An illegal variable name. Python forbids names starting with a digit, so <code>args.3d</code> can't even parse. Replaced with <code>getattr(args, "3d")</code> — returns the exact same value.
VERSION PINS environment	A few libraries needed pinning. <code>setuptools<70</code> , <code>pyarrow<16</code> , <code>datasets<3</code> so the environment installs on Kaggle — no effect on the math.

Subject fidelity collapses on hard edits

Cases split into balanced difficulty tertiles by a composite of translation, rotation, scale and mask area (our own construct, seed 42). Background stays strong; SUBC does not.

SUBC ↑ by difficulty axis 0.80–0.97



Group	SUBC↑	BGC↑	WE↓	MD↓
easy	0.942	0.971	0.046	4.85
medium	0.927	0.966	0.046	6.98
hard	0.865	0.964	0.050	13.37

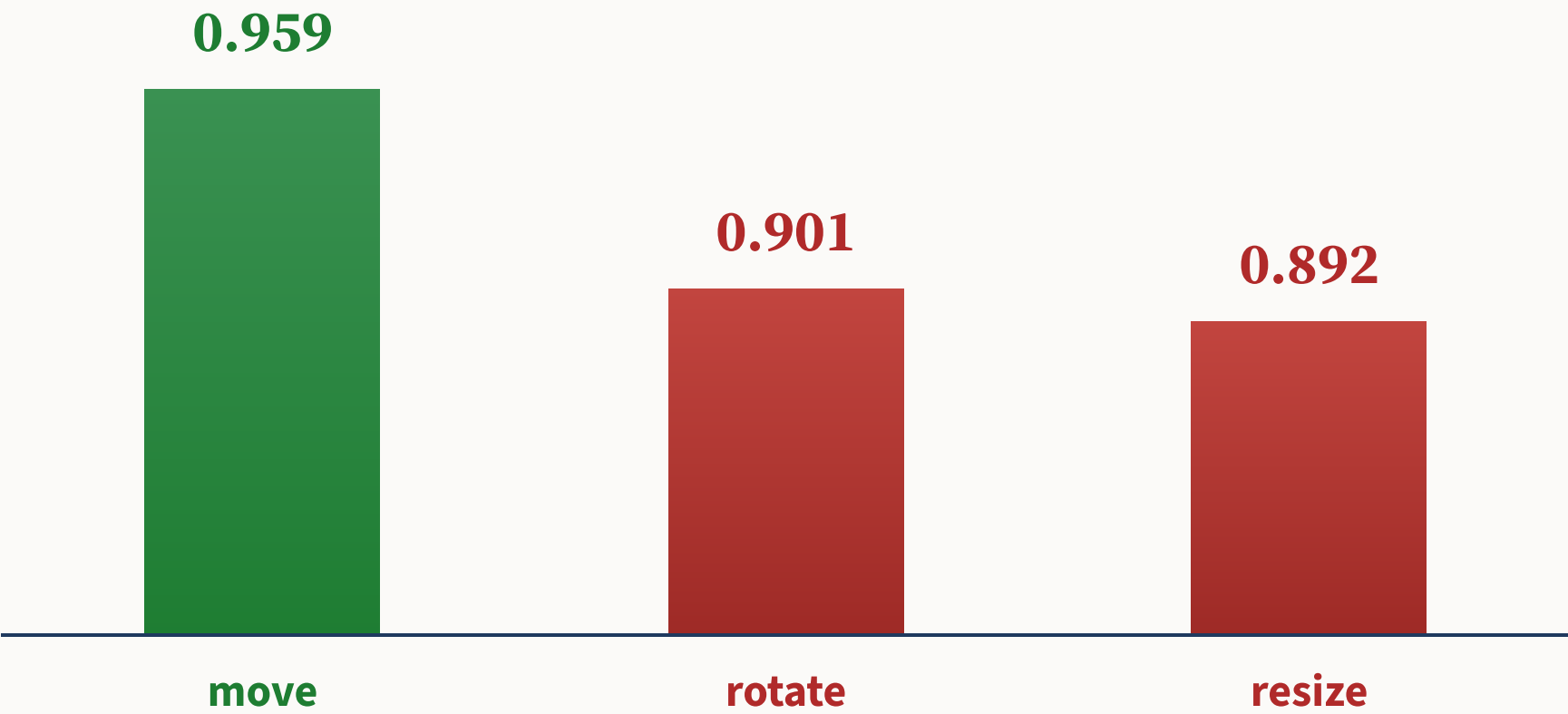
BGC stays flat (≈0.96–0.97) across all tiers. The failure is specifically **subject-structural**, concentrated in the hard tertile.

Move is solved; rotate & resize are not

Translation is near-trivial. The structural gap lives in rotation and resize — the transforms that resample the object and open larger gaps.

SUBC ↑ by edit type

axis 0.80–0.97



Type	SUBC↑	BGC↑	WE↓	MD↓
move	0.959	0.967	0.049	3.75
rotate	0.901	0.967	0.043	10.43
resize	0.892	0.967	0.049	9.96

THE TARGET

Hard rotate & hard resize — sub-group MD ≈ 15.7 and ≈ 16.5. Every experiment aims here.

PART III

Five Experiments

1 Tuning

2 Input quality

3 Identity

4 Geometry-aware TCA

5 Preservation

Can FreeFine's own controls recover it?

Question. The hard-case weakness is structural — but maybe the pipeline already exposes a knob that fixes it. Test the three documented controls before touching any code path.

CONTROL 1 · start_step

When structure injection begins

The denoising step at which TCA starts pulling structure from the inverted source. Swept 25 → 45. Earlier = more source influence.

CONTROL 2 · start_layer

Which attention layers modulate

Which U-Net self-attention layers TCA touches (4 / 8 / 12). **Fixed first:** a hardcoded range overrode this knob, making it inert.

CONTROL 3 · prompt CFG

Guidance on a text prompt

Classifier-free guidance scale (7.5 / 10) on a prompt naming the object, steering the regenerated content.

All ten variants · rotate_hard (n=22) & resize_hard (n=23)

rotate_hard (n=22)

Variant	SUBC↑	BGC↑	WE↓	MD↓
baseline	0.8523	0.9664	0.0416	14.010
ss25	0.8497	0.9681	0.0459	15.246
ss30	0.8519	0.9675	0.0437	15.911
ss40	0.8528	0.9676	0.0398	15.013
ss45	0.8529	0.9666	0.0384	15.090
14	0.8524	0.9672	0.0425	15.303
18	0.8522	0.9670	0.0424	14.329
112	0.8517	0.9669	0.0412	14.681
p7.5	0.8528	0.9675	0.0416	14.551
p10	0.8529	0.9680	0.0416	14.685

resize_hard (n=23)

Variant	SUBC↑	BGC↑	WE↓	MD↓
baseline	0.8391	0.9631	0.0585	19.471
ss25	0.8319	0.9618	0.0658	21.135
ss30	0.8376	0.9636	0.0619	20.700
ss40	0.8413	0.9637	0.0555	20.157
ss45	0.8414	0.9624	0.0535	18.089
14	0.8378	0.9622	0.0587	21.688
18	0.8382	0.9629	0.0586	21.611
112	0.8383	0.9616	0.0579	22.453
p7.5	0.8390	0.9626	0.0586	18.385
p10	0.8391	0.9625	0.0587	17.746

WHAT IMPROVES

prompt-CFG 10 cuts resize-hard **MD 19.47→17.75**; start_step 45 also helps (18.09) and gives best rotate SUBC (0.8529) + lowest rotate WE (0.0384).

WHAT DOESN'T

Low start_step hurts (ss25 resize MD 21.1); start_layer worsens it (l12 22.5). **No control raises SUBC** — it stays 0.8517–0.8529 (rotate).

OVERALL READ

Controls trade a little precision but can't move structure — and SUBC swings are within the **n=22/23 noise band**. Structural fidelity isn't a tuning knob.

EXPERIMENT 1 · FULL RESULTS – MOVE & WHOLE SUBSET

move_all (n=67) & all_200 (n=200, with FID family)

move_all (n=67)

Variant	SUBC↑	BGC↑	WE↓
baseline	0.9590	0.9639	0.0495
ss25	0.9542	0.9647	0.0534
ss30	0.9571	0.9642	0.0512
ss40	0.9605	0.9627	0.0480
ss45	0.9612	0.9629	0.0468
14	0.9587	0.9639	0.0497
18	0.9587	0.9639	0.0496
112	0.9590	0.9628	0.0494
p7.5	0.9588	0.9638	0.0496
p10	0.9587	0.9639	0.0497

all_200 (n=200) · FID family subset-inflated, trend only

Variant	SUBC↑	BGC↑	WE↓	FID↓	FDD↓	KD↓
baseline	0.9154	0.9657	0.0488	132.28	1636.8	0.1285
ss25	0.9113	0.9667	0.0540	132.57	1636.1	0.1270
ss30	0.9139	0.9663	0.0513	132.73	1636.6	0.1264
ss40	0.9166	0.9650	0.0468	133.09	1640.3	0.1347
ss45	0.9174	0.9641	0.0452	133.58	1643.7	0.1397
14	0.9152	0.9658	0.0492	132.41	1636.9	0.1264
18	0.9152	0.9658	0.0491	132.50	1637.2	0.1303
112	0.9152	0.9650	0.0486	132.82	1637.5	0.1276
p7.5	0.9154	0.9656	0.0489	132.35	1637.8	0.1273
p10	0.9154	0.9657	0.0490	132.51	1637.9	0.1326

WHAT IMPROVES

Move is near-ceiling and stable; ss45 nudges it to **0.9612** and gives the best all_200 SUBC (0.9174) with lower WE (0.0452).

THE TRADE-OFF

That same ss45 SUBC gain comes with **higher FID/FDD** (133.6 / 1643.7); low start_step lowers move (ss25 0.9542). FID-family is subset-noisy — trend only.

OVERALL READ

No single config both lifts SUBC *and* holds realism. Tuning is now fully characterised — a clean baseline for the architectural experiments. (Empty-prompt CFG ≡ baseline.)

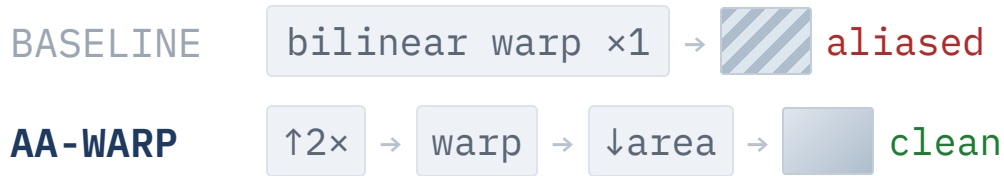
A cleaner coarse, and more structure injected

Question. Two upstream possibilities for the hard-case gap: the coarse image is too aliased to refine well, or the structure injection stops too early. Test each directly.

LEVER 1 · AA-Warp

Anti-aliased coarse composite

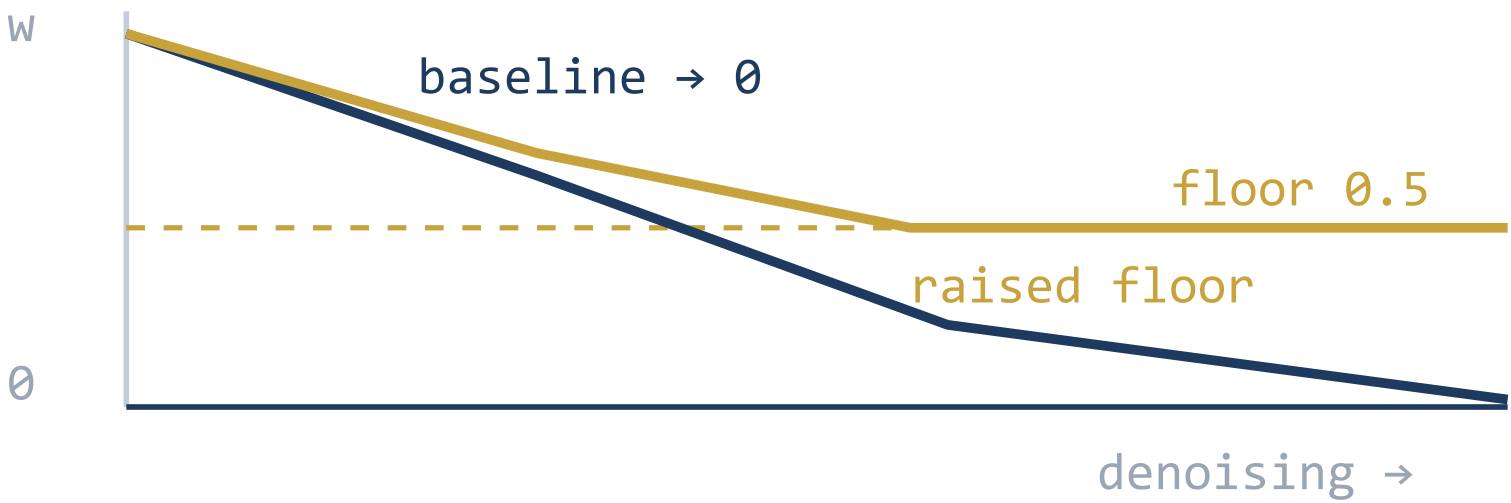
Replace the single-pass **bilinear** warp — which aliases hard edges — with a **supersampled-Lanczos** warp, so refinement starts from a cleaner image.



LEVER 2 · Adaptive-ES

Raise the injection floor

end_scale is the floor the structure-injection weight decays to — baseline 0. Raise it (fixed 0.25 / 0.5, or difficulty-adaptive) so source structure persists later into denoising.



AA-Warp & Adaptive-ES results

rotate_hard (n=22)

Variant	SUBC↑	BGC↑	WE↓	MD↓
baseline	0.8523	0.9664	0.0416	14.010
aaw	0.8537	0.9663	0.0447	14.544
es25	0.8524	0.9671	0.0423	14.736
es50	0.8522	0.9672	0.0434	14.964
esA	0.8522	0.9672	0.0434	14.964
aaw_esA	0.8536	0.9662	0.0463	15.018

resize_hard (n=23)

Variant	SUBC↑	BGC↑	WE↓	MD↓
baseline	0.8391	0.9631	0.0585	19.471
aaw	0.8403	0.9628	0.0573	19.666
es25	0.8388	0.9630	0.0591	22.059
es50	0.8380	0.9632	0.0600	24.452
esA	0.8380	0.9632	0.0600	24.452
aaw_esA	0.8388	0.9631	0.0590	20.297

end_scale: clean negative. Raising the floor degrades monotonically (resize-hard MD 19.5 → 22.1 → 24.5). Baseline decay-to-0 is already optimal.

move_all (n=67)

Variant	SUBC↑	BGC↑	WE↓
baseline	0.9590	0.9639	0.0495
aaw	0.9592	0.9635	0.0454
es25	0.9588	0.9644	0.0496
es50	0.9582	0.9652	0.0500
esA	0.9583	0.9646	0.0499
aaw_esA	0.9586	0.9647	0.0459

all_200 (n=200) • FID family trend-only

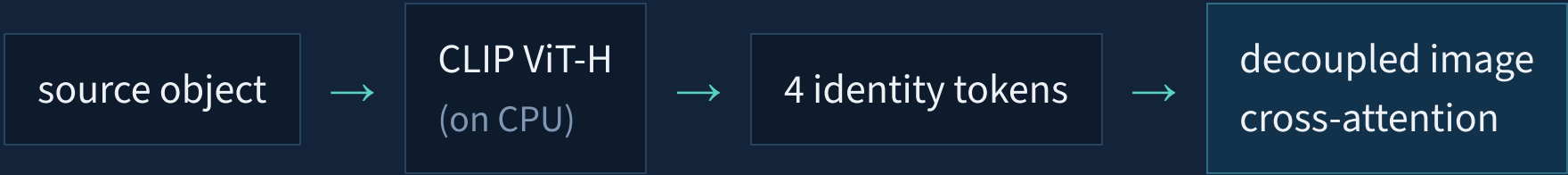
Variant	SUBC↑	BGC↑	WE↓	FID↓	FDD↓
baseline	0.9154	0.9657	0.0488	132.28	1636.8
aaw	0.9156	0.9654	0.0486	132.36	1637.3
es25	0.9155	0.9660	0.0492	132.31	1637.9
es50	0.9152	0.9666	0.0498	132.52	1638.5
esA	0.9151	0.9661	0.0494	132.55	1638.0
aaw_esA	0.9153	0.9659	0.0492	132.63	1638.4

AA-Warp: hygiene, not a fix. Tiny SUBC nudge (+0.0014 rot), slightly cleaner WE — no movement on the bottleneck; all_200 flat.

Inject identity through a rotation-tolerant channel

Hypothesis. TCA matches features by spatial similarity, so a strongly rotated object no longer matches its source and loses identity. A CLIP image embedding is largely orientation-invariant — feed identity through that instead.

PIPELINE ADDITION



The IP-Adapter branch adds its masked output to the refinement U-Net's text cross-attention, scaled by the adapter strength. Identity arrives through a channel TCA does not have.

SWEEP

Adapter scale **0.5 / 0.8 / 1.1 / 1.4** (ip50–ip140), masked to the target object only.

ENGINEERING

CLIP encoder placed on CPU; ~84 min/variant — the memory-vs-time trade that kept it inside the T4 budget.

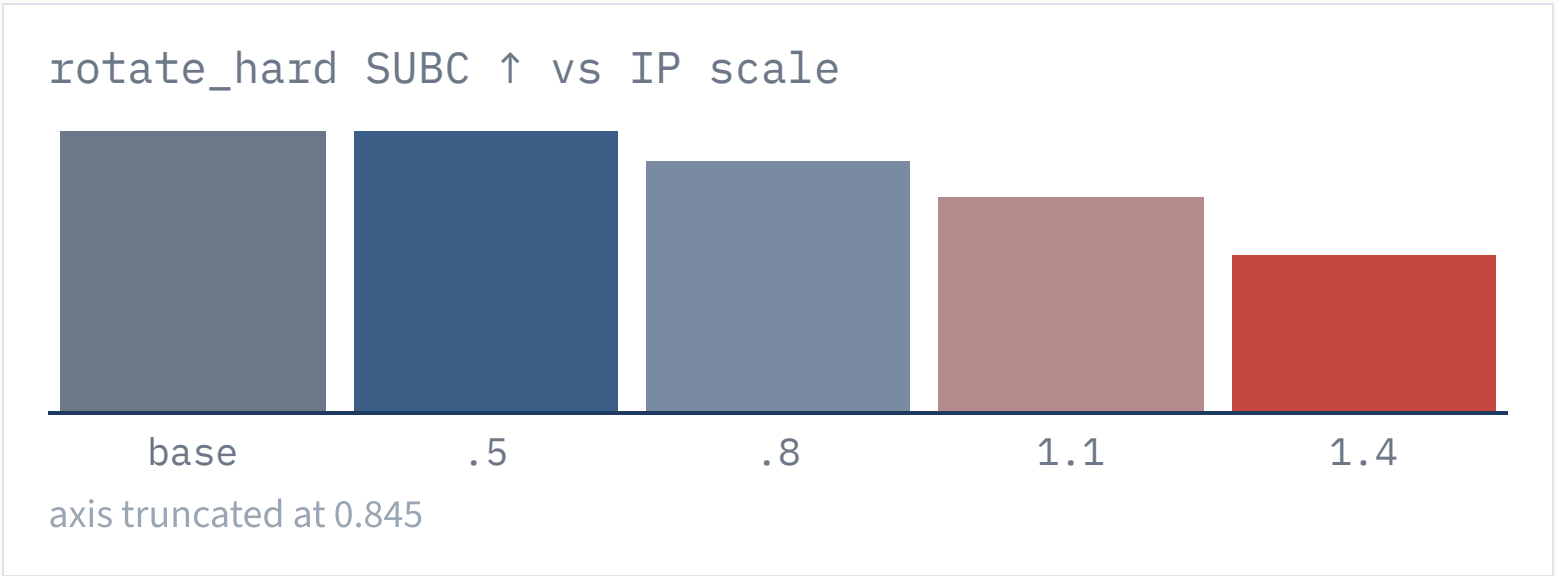
Identity conditioning degrades SUBC monotonically

rotate_hard (n=22) · resize_hard (n=23)

Var	rot SUBC↑	rot MD↓	res SUBC↑	res MD↓
base	0.8523	14.010	0.8391	19.471
ip50	0.8523	14.675	0.8396	21.486
ip80	0.8518	14.862	0.8394	18.753
ip110	0.8512	15.467	0.8393	19.429
ip140	0.8502	15.228	0.8390	21.540

move_all (n=67) · all_200 (n=200)

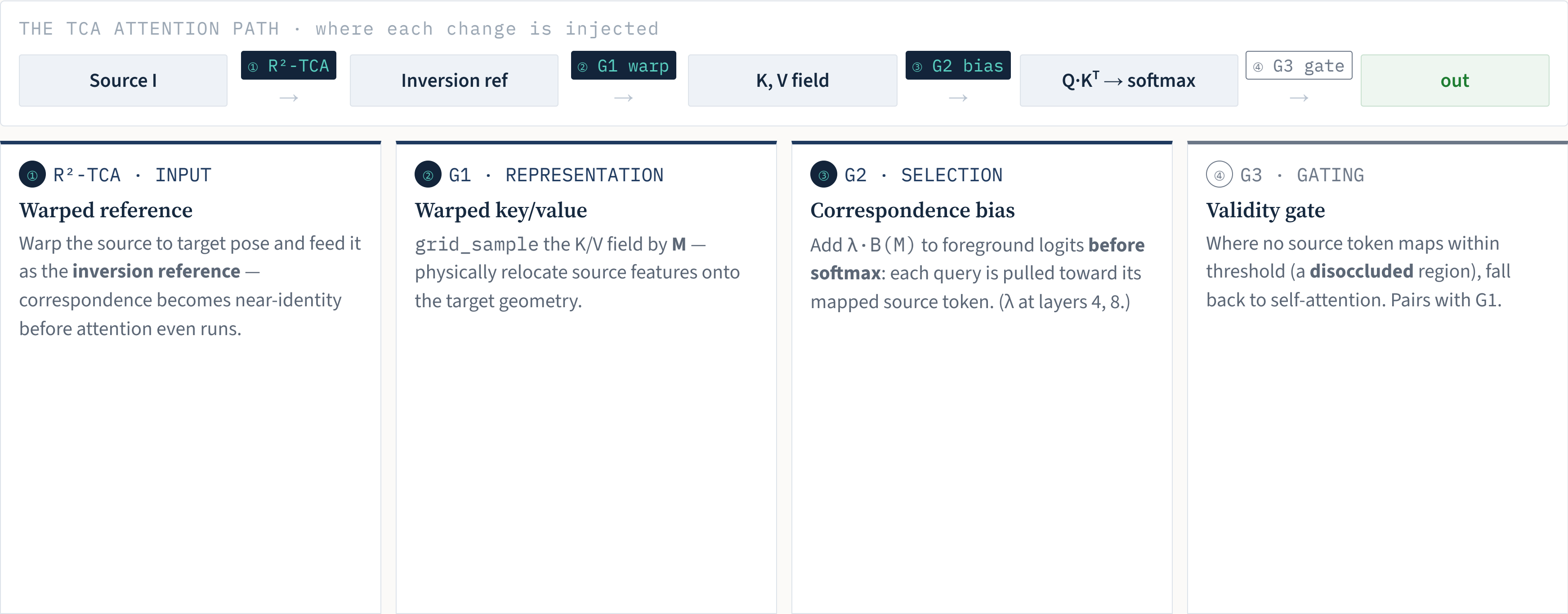
Var	mv SUBC↑	200 SUBC↑	200 BGC↑	200 FID↓
base	0.9590	0.9154	0.9657	132.28
ip50	0.9590	0.9154	0.9656	132.08
ip80	0.9589	0.9153	0.9656	132.11
ip110	0.9587	0.9150	0.9656	132.15
ip140	0.9585	0.9147	0.9655	132.27



The 4-token adapter injects **coarse semantic identity, not fine geometric structure**. Wrong problem — hypothesis ruled out.

Put the known affine inside the attention

Code-level diagnosis: the TCA query attends to source K/V through a column mask only — it knows *which* tokens are foreground, never *where* they map. The per-edit affine **M** is known exactly, yet never used inside attention; correspondence is left to feature similarity, which fails under rotation/scale.



Every level lands at or below baseline · rot_h (n=22) & res_h (n=23)

rotate_hard (n=22)

Variant	SUBC↑	BGC↑	WE↓	MD↓
baseline	0.8523	0.9664	0.0416	14.010
g2_l4	0.8318	0.9596	0.0484	14.224
g2_l8	0.8269	0.9575	0.0496	14.600
g1_warp	0.8438	0.9611	0.0422	15.419
g1warp_gate	0.8445	0.9657	0.0419	15.297
r2 (R²-TCA)	0.8495	0.9629	0.0399	14.856

resize_hard (n=23)

Variant	SUBC↑	BGC↑	WE↓	MD↓
baseline	0.8391	0.9631	0.0585	19.471
g2_l4	0.8284	0.9561	0.0624	17.907
g2_l8	0.8257	0.9541	0.0632	19.974
g1_warp	0.8329	0.9573	0.0602	22.283
g1warp_gate	0.8336	0.9571	0.0583	26.522
r2 (R²-TCA)	0.8334	0.9557	0.0600	21.016

G2 · LOGIT BIAS
Largest SUBC drop, and it pulls **BGC down too** — biasing logits perturbs beyond the object.

G1 · WARPED K/V
Gentler on SUBC but **inflates MD** — resampling keys displaces keypoints (resize 26.5).

R² - TCA
Closest to baseline; even **improves rotate WE** (0.0399) — yet still no SUBC gain.

Every variant lowers SUBC — and that is the clue

Variant	rot_h SUBC↑	res_h SUBC↑	move SUBC↑	200 SUBC↑	200 BGC↑	200 FID↓
baseline	0.8523	0.8391	0.9590	0.9154	0.9657	132.28
g2_l4	0.8318	0.8284	0.9415	0.9013	0.9607	135.15
g2_l8	0.8269	0.8257	0.9373	0.8979	0.9585	135.81
g1_warp	0.8438	0.8329	0.9542	0.9095	0.9598	133.47
g1warp_gate	0.8445	0.8336	0.9562	0.9106	0.9614	133.58
r2 (R²-TCA)	0.8495	0.8334	0.9560	0.9128	0.9633	132.52

All SUBC values below baseline. MD (rot_h): g1_warp 15.42, g1warp_gate 15.30, r2 14.86 • (res_h): g1warp_gate 26.52 worst.

THE SMOKING GUN
Pure translation is trivial — baseline solved it (0.959) — yet **every variant damages move too** (down to 0.937). With no rotation/scale to help, the loss is the mechanisms' own artifacts, not geometry difficulty.

THEORY CORRECTION
R²-TCA gave a geometrically-ideal reference and still didn't beat baseline. The reference pose-mismatch is **not** the bottleneck — the collapse lives downstream, in the **refinement itself**.

Anchor the object to the correct coarse

From Exp 4: the bottleneck is the refinement degrading an object the coarse already had correct. So preserve more of the coarse — and measure how much the refinement was costing.

LATENT-PRESERVATION BLEND · per denoising step

```
latent = latent · (1 - w · m)
        + coarse · (w · m)
```

```
m = object mask · w = blend strength
```

Blend the edit latent toward the coarse's own inversion latent, inside the object region. **w=0** $\equiv$ baseline; **w**→**1** pins the object to the coarse — the perfect in-plane warp.

No extra model calls. The coarse inversion trajectory is already produced by the pipeline — the blend rides on it for free.

A lever **and** a measurement: how much was the refinement costing the object the coarse already had right? Swept **w = 0.3 / 0.5 / 0.7 / 0.9**.

A first gain everywhere — that reveals a ceiling

Variant	rot_h SUBC↑	res_h SUBC↑	move SUBC↑	200 SUBC↑	200 WE↓
baseline	0.8523	0.8391	0.9590	0.9154	0.0488
pres_w03	0.8536	0.8409	0.9610	0.9175	0.0454
pres_w05	0.8535	0.8404	0.9612	0.9174	0.0453
pres_w07	0.8535	0.8400	0.9614	0.9174	0.0453
pres_w09	0.8535	0.8399	0.9614	0.9174	0.0453

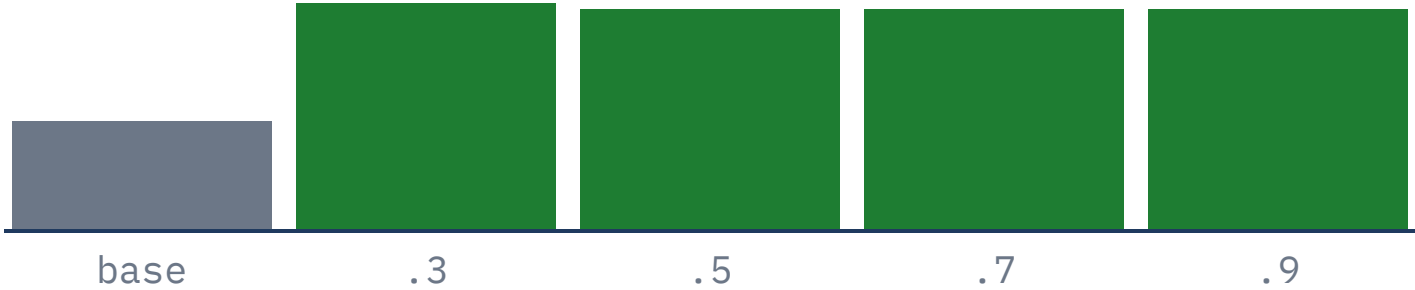
SUBC flat across w 0.3–0.9 → dose-saturated. FID rises slightly (132.3→133.5); BGC dips ~0.003. MD within hard-group noise.

First positive result. SUBC beats baseline in every group (+0.0021 on all_200), WE −7%, at a small FID cost.

≈99.9%

of the perfect-warp subject-consistency ceiling, already reached by the refinement.

all_200 SUBC ↑ vs w (axis 0.910–0.920)



The hard-case collapse is present in the **perfect warp itself** — intrinsic to the 2D transform and metric view-sensitivity, not an algorithmic deficiency.

PART IV

Engineering & Direction

The compute behind the study — and where the work goes next.

Engineering under a 12-hour wall

Kaggle, 2×NVIDIA T4 (~15 GB each), hard 12-hour per-commit limit — every run is deadline-guarded and sharded.

≈35 s per case, 1 GPU (≈66 s at 2 GPUs — attention contention).

70–85 m to generate one balanced-200 variant.

2 envs generation (torch 2.1.1 / diffusers 0.18) vs metrics (torch 2.6); MD seeded 42 everywhere.

PER-COMMIT RUNTIME (T4×2)

Phase	Generation	Total
Phase 2/2b	5 variants × ≈68 m	≈10.5 h
B1	5 × ≈72 m (≈6.0 h)	≈9 h
B2	4 × ≈84 m (IP enc. on CPU)	≈8.5 h
B3	5 × ≈73 m (≈6.1 h)	≈9.5 h
B4	4 × ≈76 m (≈5.0 h)	≈8.5 h

Sweeps are per-variant isolated and deadline-guarded — a single failure never costs a whole commit.

NEXT STEPS

Where the real headroom now lies

2D in-plane subject-consistency is at its ceiling — so the open problems are **out-of-plane geometry** and **scene effects**, neither addressable inside the current 2D pipeline.

DIRECTION A

3D / out-of-plane editing

Out-of-plane rotation reveals surfaces a 2D affine warp can never produce — FreeFine's own **GeoBench-3D** track is wide open.

Plan: replace the geometric-transform stage with generative novel-view synthesis (Zero123++ / SV3D) to hallucinate the unseen side, then refine with the same training-free TCA.

DIRECTION B

Effects-aware editing

The object moves but its **shadow, reflection & contact shading** stay behind — a glaring realism gap none of the seven consistency metrics even measure.

Plan: synthesize the object's cast effects at the *target* and remove them at the *source* — and design an evaluation that actually scores them.